

Azure–AWS Site-to-Site VPN Configuration

“Implementation of Secure Multi-Cloud Connectivity Between Microsoft Azure and AWS Using Site-to-Site VPN”

Aim

To design, configure, and validate a secure Site-to-Site VPN connection between Microsoft Azure and Amazon Web Services (AWS), enabling seamless communication, secure data transfer, and resource accessibility across two different cloud platforms while maintaining confidentiality, integrity, and network reliability.

Project Overview

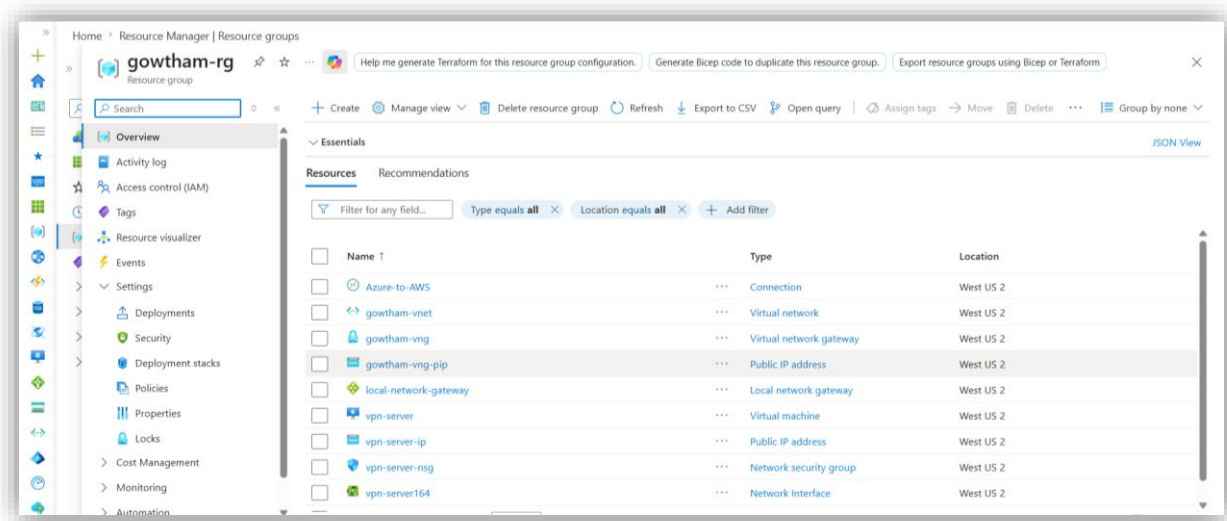
This project focuses on establishing a secure Site-to-Site VPN tunnel between Microsoft Azure and Amazon Web Services (AWS), two of the most widely used cloud computing platforms. The implementation enables private communication between resources deployed in an Azure Virtual Network (VNet) and an AWS Virtual Private Cloud (VPC) without exposing traffic to the public internet.

Objectives

- Configure networking components in Azure and AWS.
- Establish an IPsec Site-to-Site VPN tunnel between Azure VNet and AWS VPC.
- Enable secure communication between cloud-hosted resources.
- Verify connectivity through testing and monitoring.
- Demonstrate a practical multi-cloud networking solution.
- Ensure secure and encrypted data transmission across cloud platforms.

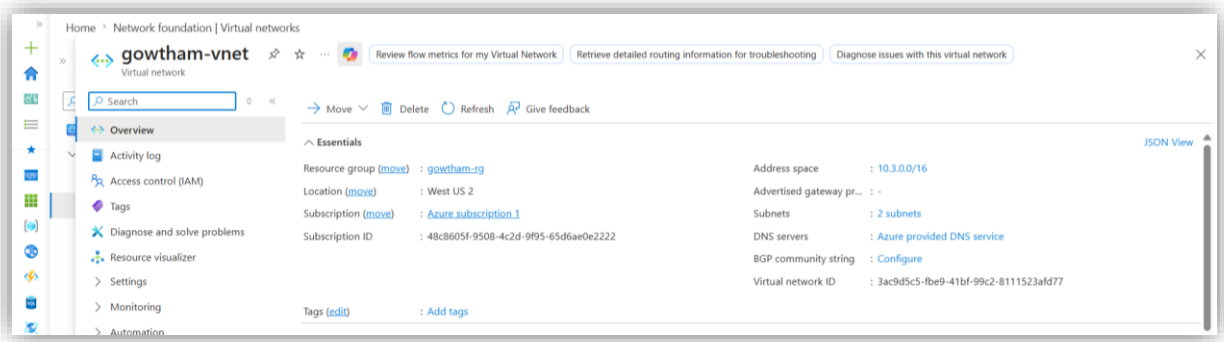
Infrastructure in Azure

- Create Resource Group: gowtham-rg

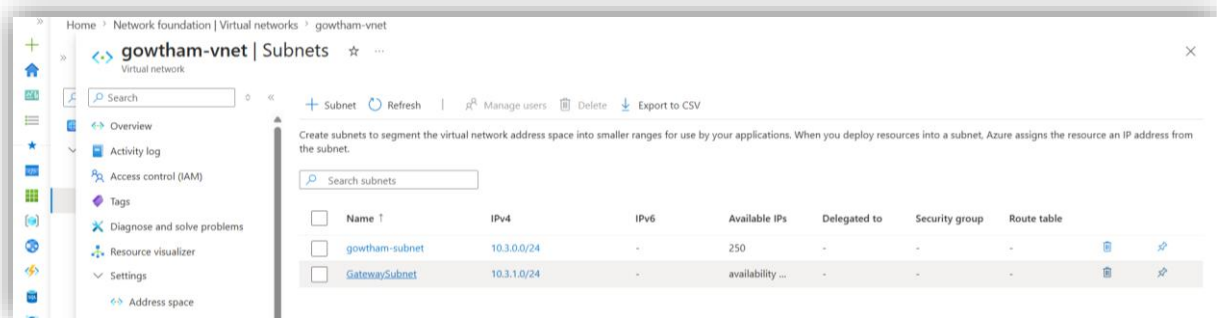


- Create Vnet: gowtham-vnet

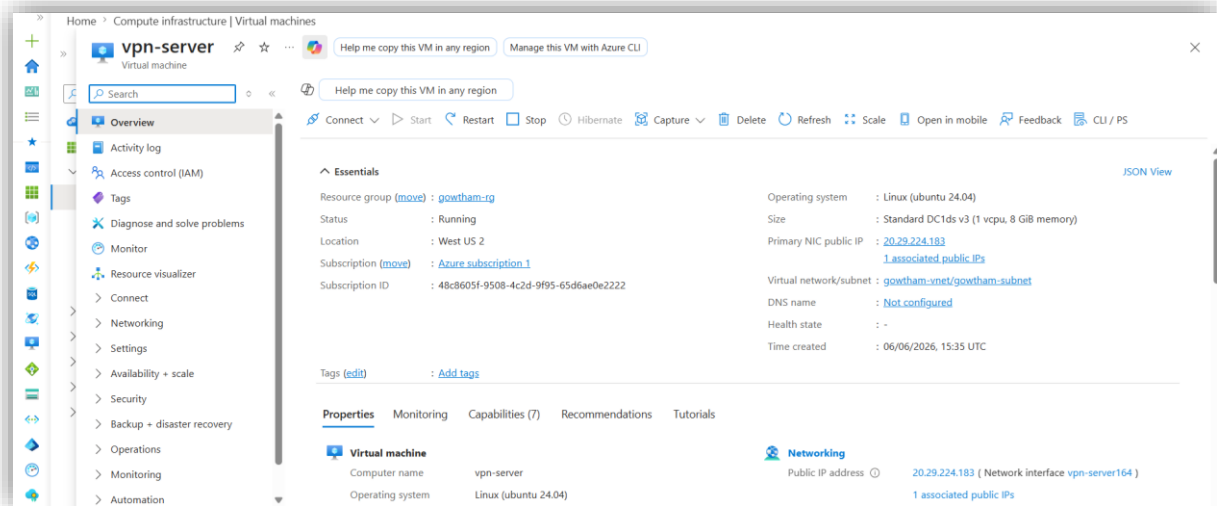
Azure–AWS Site-to-Site VPN Configuration



- Create Two Subnets in Vnet, One Dedicated Subnet For gateway : GatewaySubnet and Another Subnet for vm



- Create a Virtual Machine: vpn-server

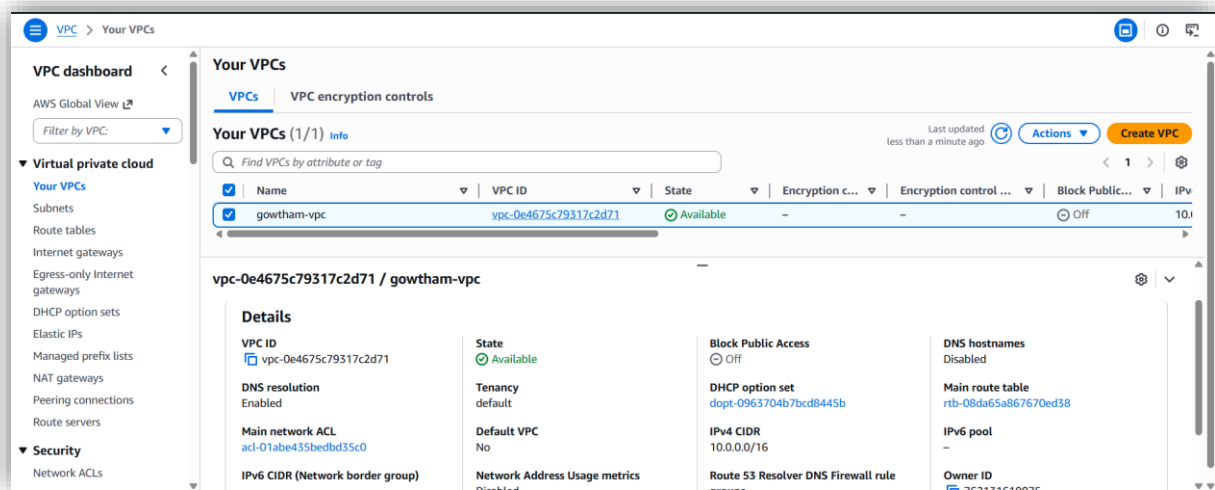


- Building Infrastructure in Azure Is Completed

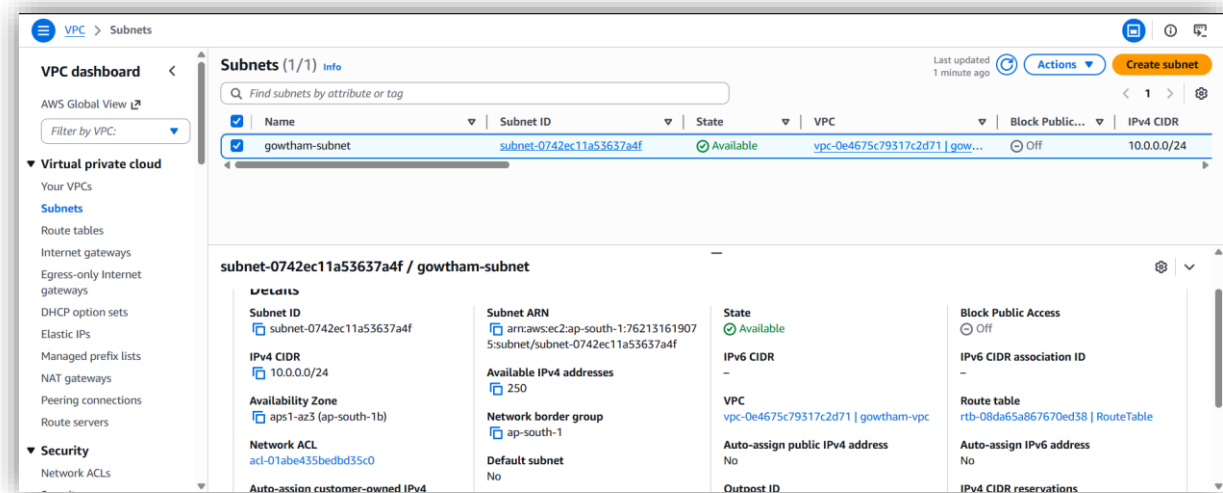
Infrastructure in AWS

- Create VPC: gowtham-vpc

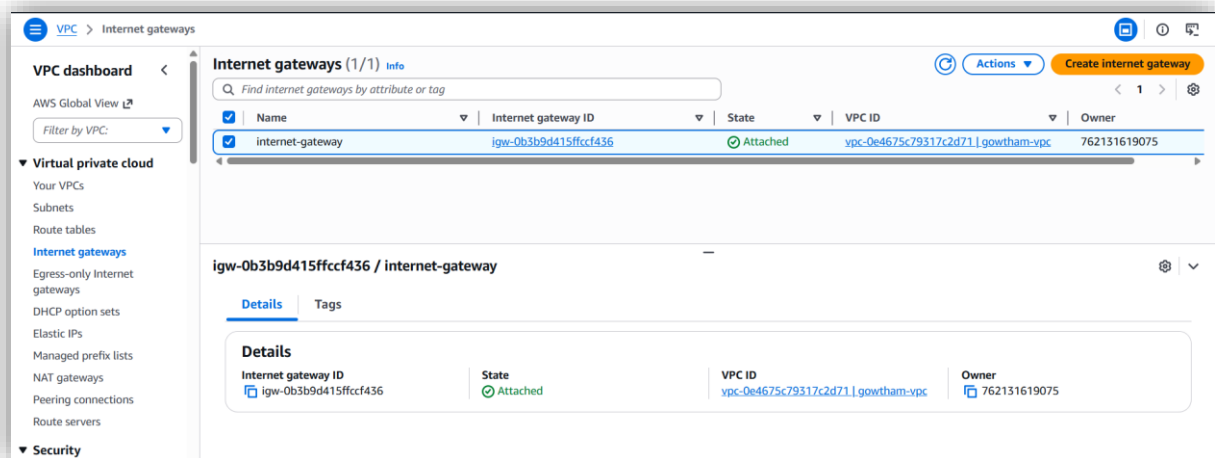
Azure–AWS Site-to-Site VPN Configuration



- Create subnet: gowtham-subnet

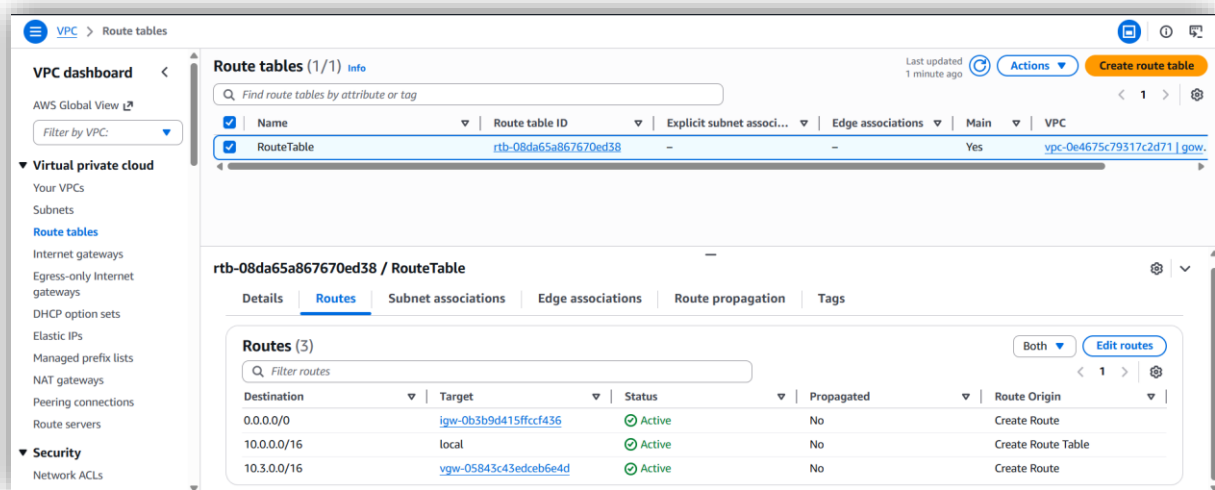


- Create Internet Gateway and attach to VPC

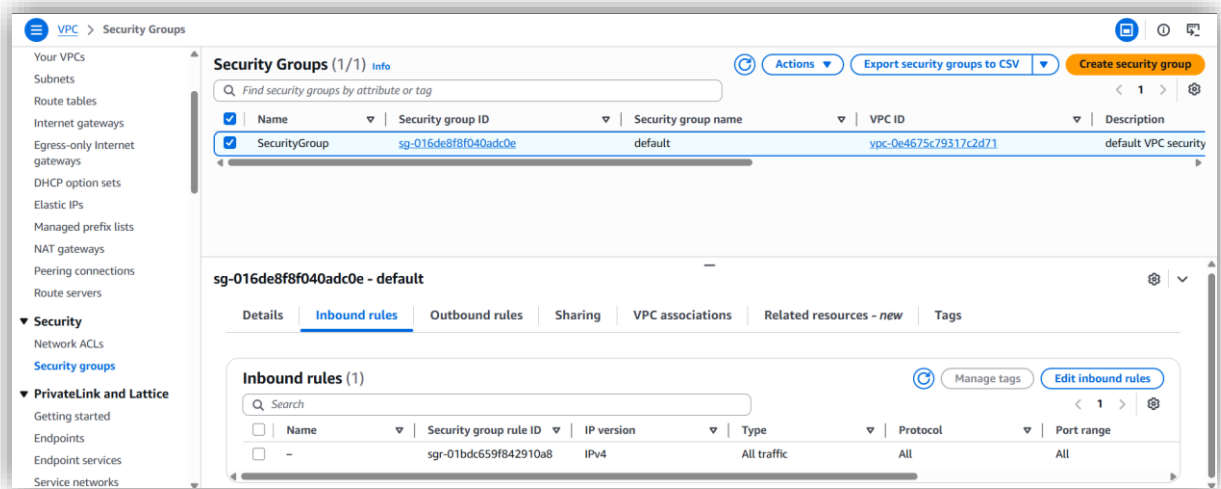


- Rename Route Table and edit Routes:

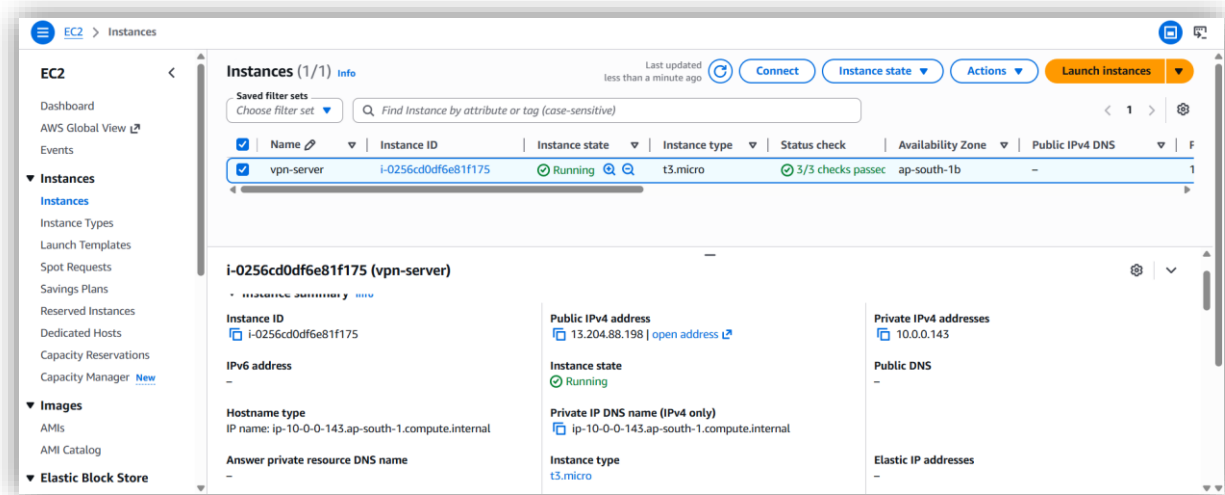
Azure–AWS Site-to-Site VPN Configuration



- Rename Security Group and edit Rules:



- Create EC2:

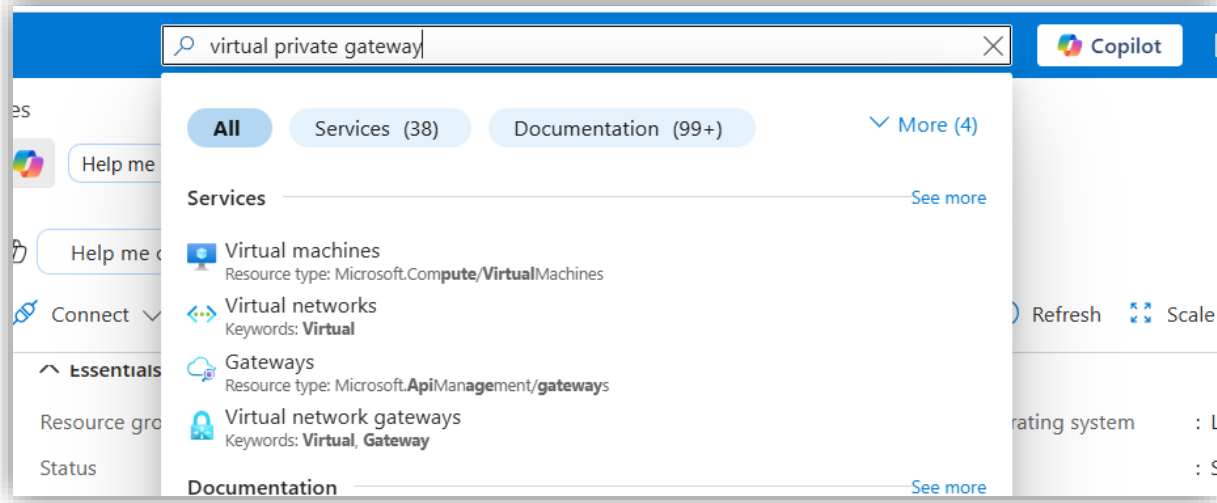


VPN SETUP

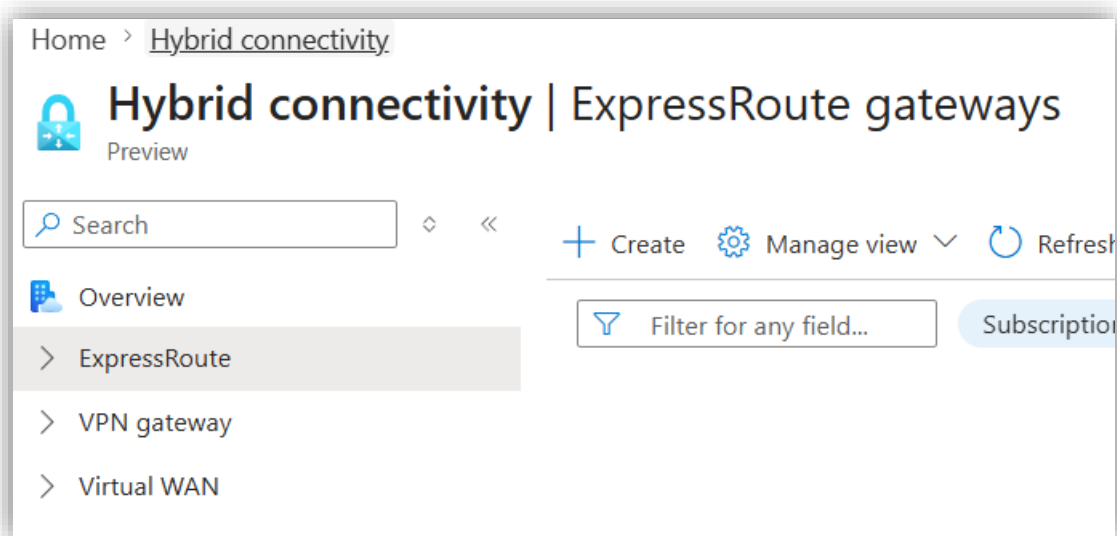
➔ Start Building VPN setup in Azure Virtual Network – VPN Gateway

Azure–AWS Site-to-Site VPN Configuration

- As it is used in AWS VPN setup
- In Azure, type in search bar Virtual Private gateway complete else it won't show

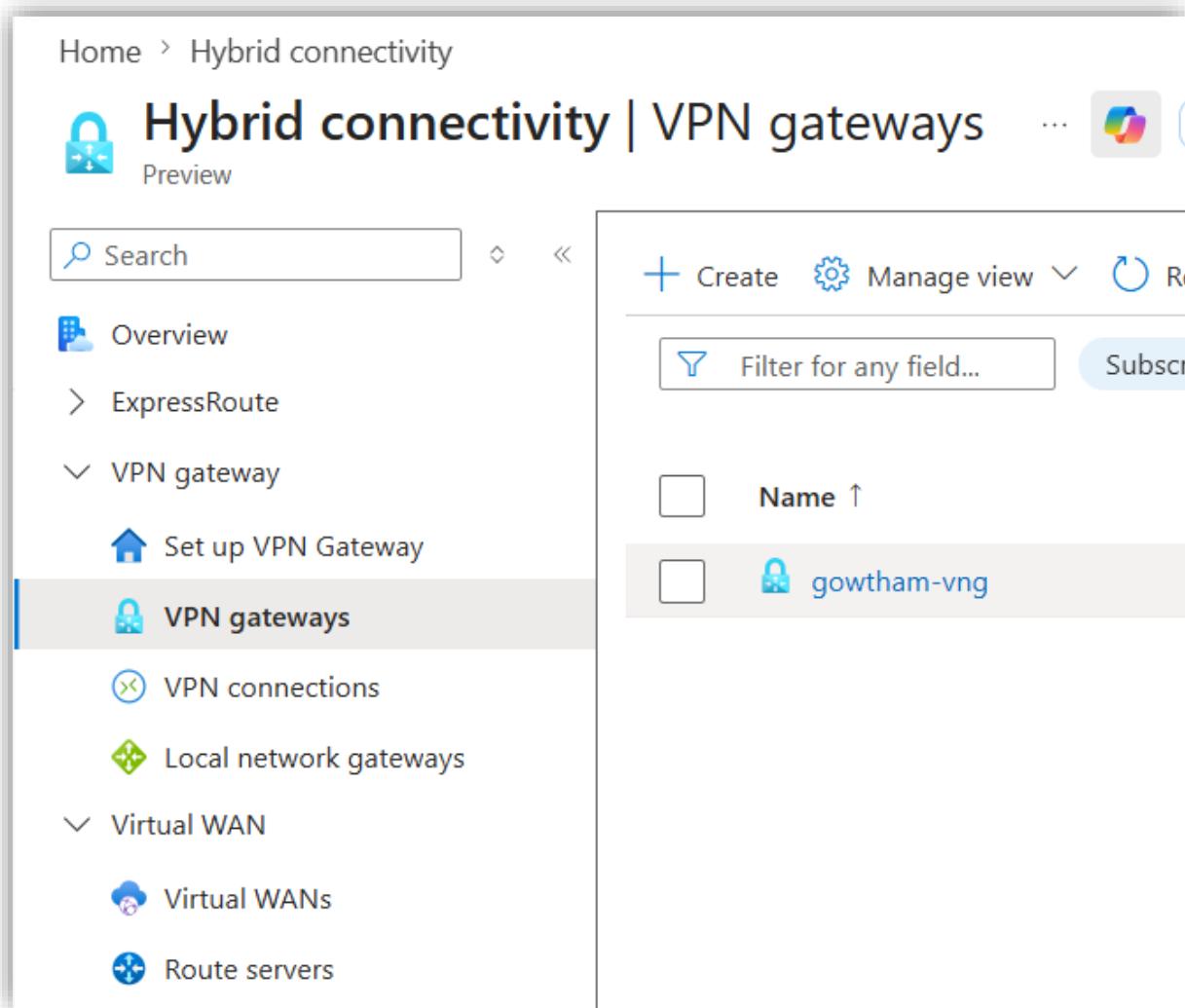


- Select Virtual Network Gateways



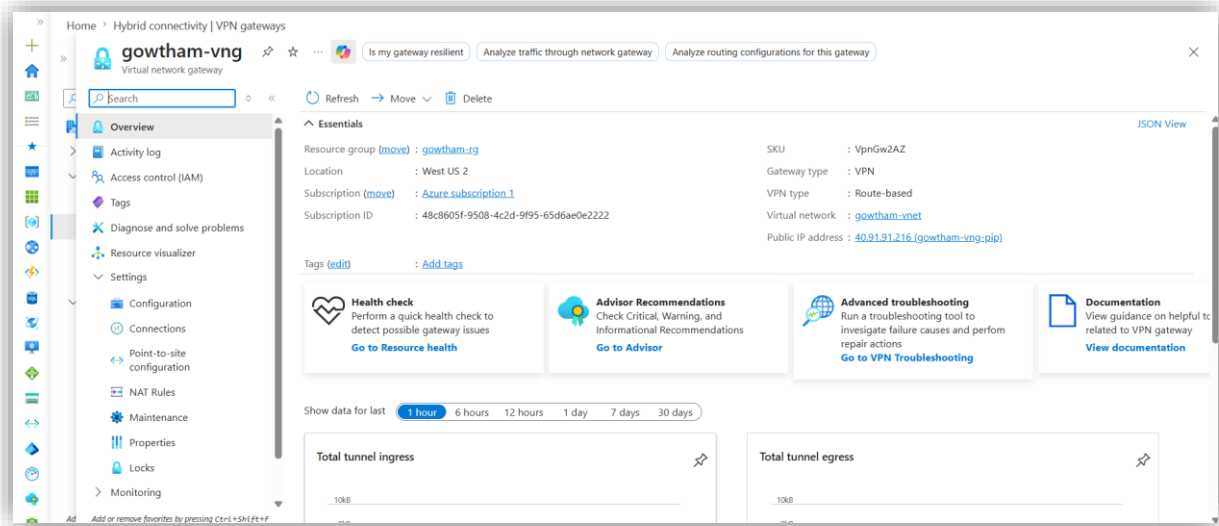
- From left bar select VPN gateway -> VPN gateway

Azure–AWS Site-to-Site VPN Configuration



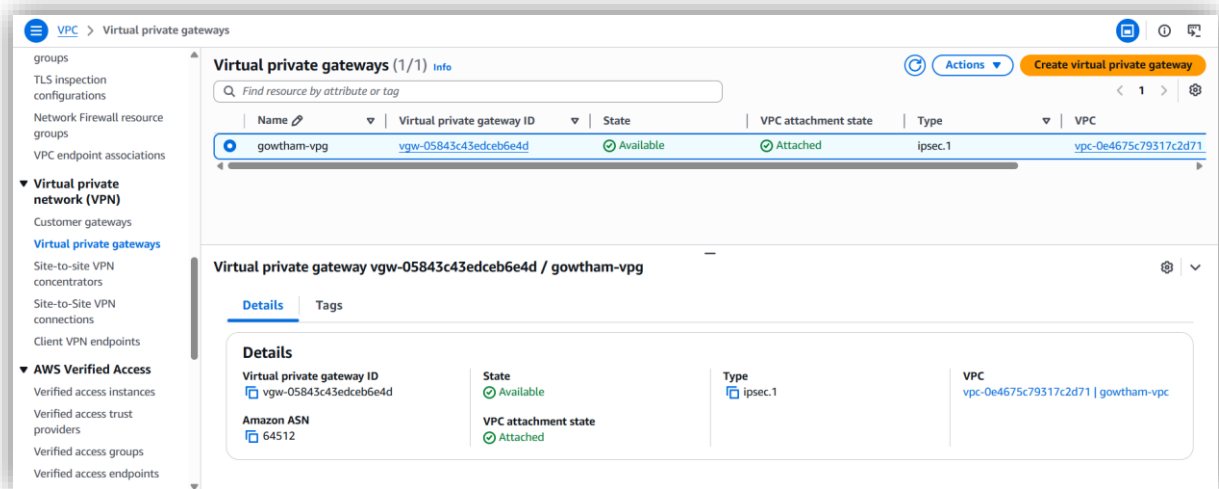
- Click on create
- Basic
 - Name: any
 - Region: West US
- Gateway Configuration:
 - Gateway Type: VPN
 - SKU: VpnGw2AZ
 - Virtual Network: select yours
 - Subnet: Default selected
- If Gatewaysubnet is not created before, it shows error
- Public IP:
 - Public IP: Create new
 - Public ip address name: any
 - Enable active-active mode: Disabled
 - Configure BGP: Disabled
- Review and Create -> create
- Takes time to create 10 to 15 minutes

Azure–AWS Site-to-Site VPN Configuration



AWS VPN Setup

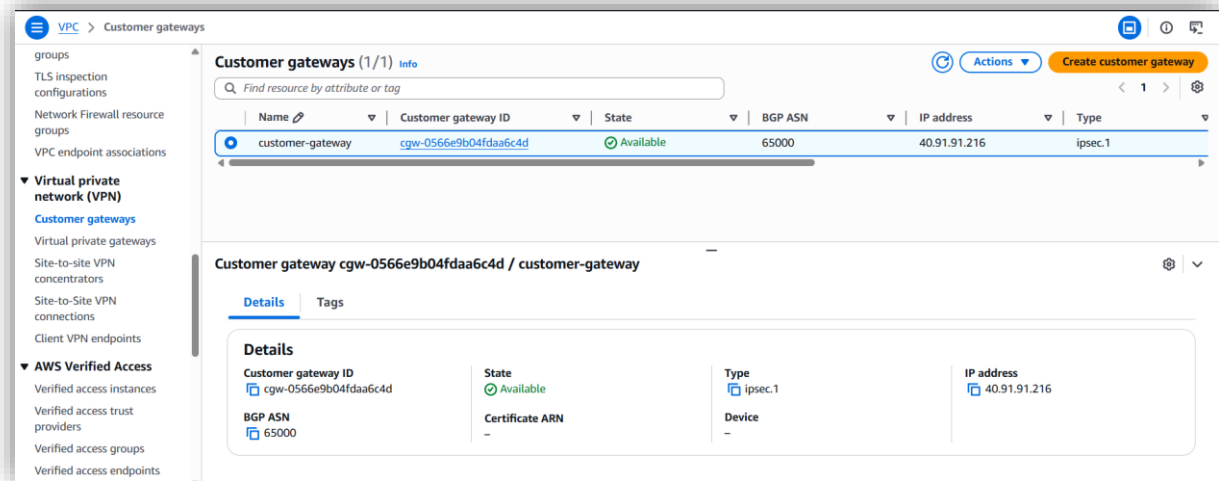
- From VPC left bar
- Under Virtual Private network (VPN)
- You can find **Virtual Private Gateways** need for VPN
- Name: any
- Create
- After creating VPN attach that VPN to VPC



Customer Gateway

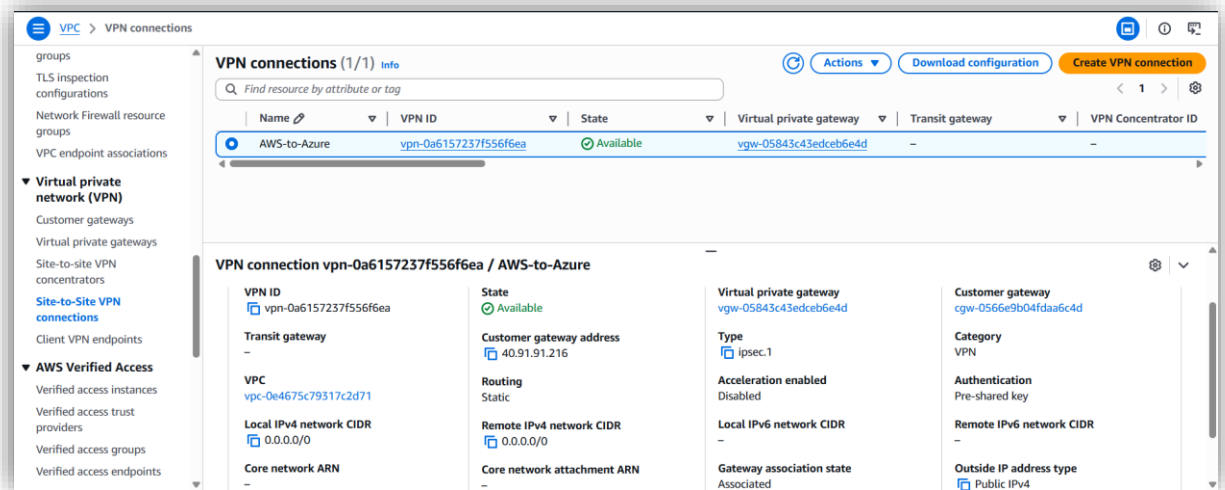
- From Left bar Under Virtual Private Network
- Select Customer gateway
- Click on **Create Customer Gateway**
- Fill Details:
 - Name tag: any
 - BGP ASN: 65000 (default)
 - Ip address: provide azure vpn public ip
 - Certificate ARN: leave it
 - Device optional: choice
 - Create gateway
- After creating Customer Gateway

Azure–AWS Site-to-Site VPN Configuration



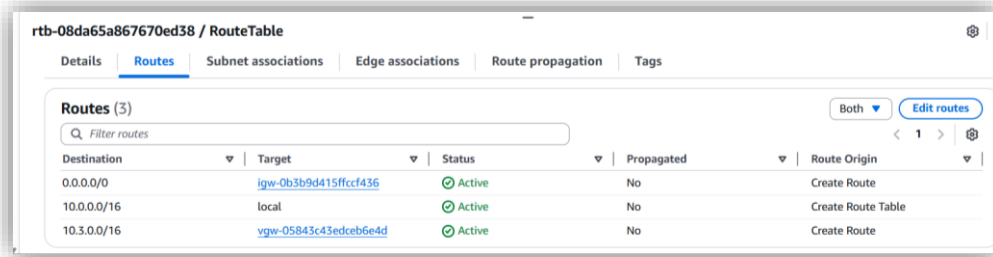
Now SITE TO SITE VPN CONNECTIONS

- From left bar under Virtual Private Network (VPN)
- Select site to site VPN connections
- Create VPN connection
- Fill details:
 - Name tag: any
 - Target gateway type: select virtual private gateway
 - Virtual private gateway: select id
 - Customer gateway: existing
 - Customer gateway id: select created id
 - Routing Options: static
 - Routing IP prefix: Azure VM's Vnet range
- After creating VPN connection

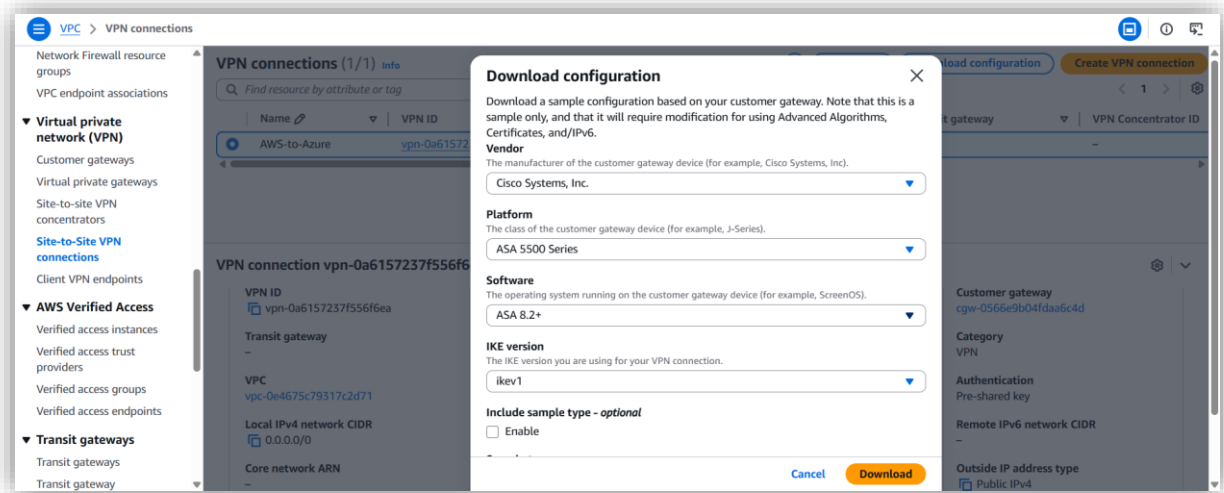


- After creating VPN connection
- Go to Route table and write rules
 - Destination: 10.3.0.0/16 (Azure vnet ip range) | Target: Virtual Private Gateway | Tag: VPGid,...

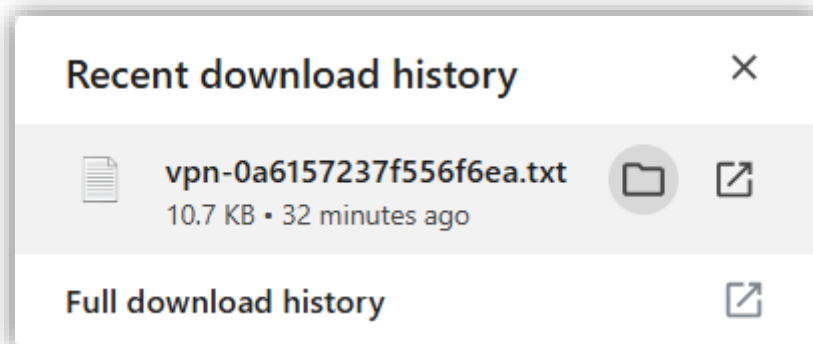
Azure–AWS Site-to-Site VPN Configuration



- Click on site to site connections and select connection
- Download configuration gets active, select it
- Select details and download
- Vendor: Cisco Systems, inc
- Platform: ASA 5500 Series
- ASA: 8.2+
- IKE version: ikev1
- Select download, a notepad is downloaded



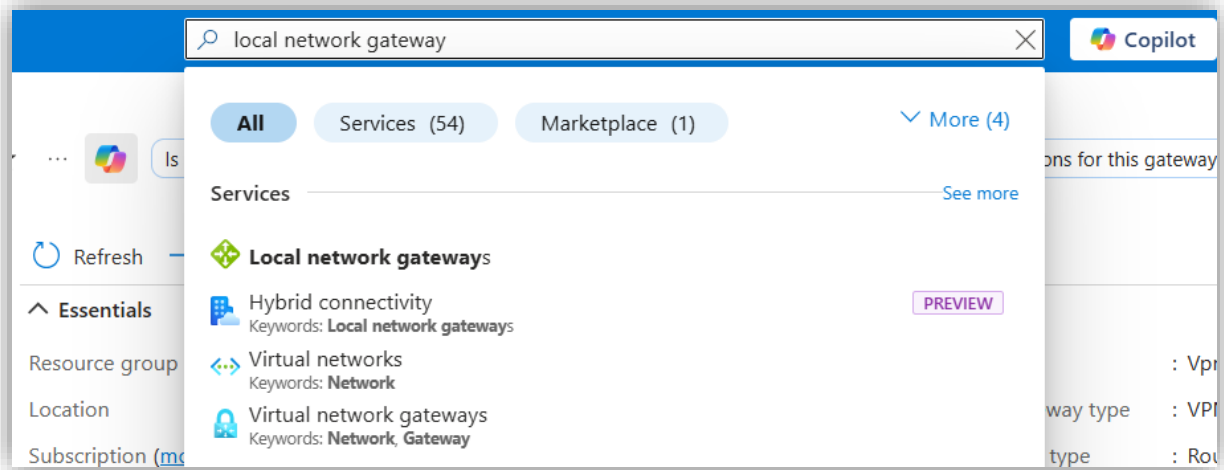
- Select download, a notepad is downloaded



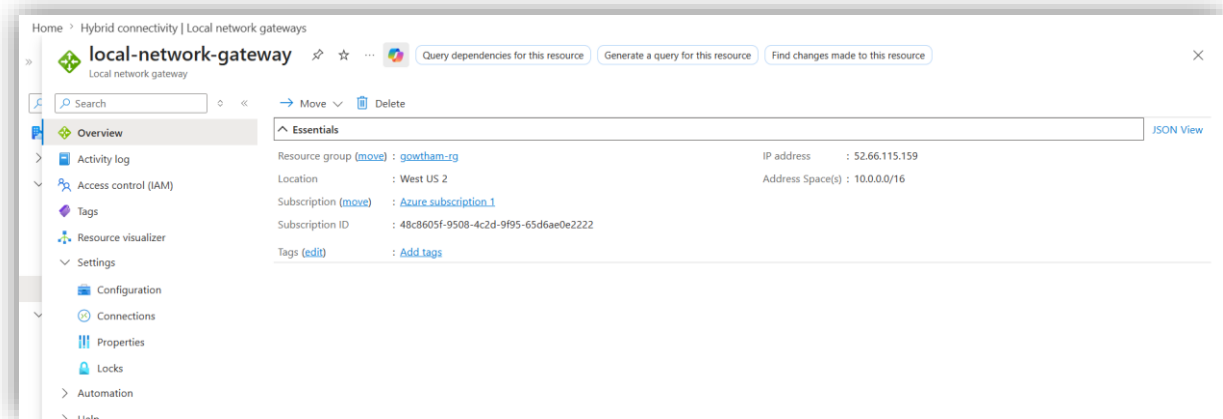
From **Azure Setup Open VPN gateway** and create **connection**

- Search Local Network Gateway and select it

Azure–AWS Site-to-Site VPN Configuration



- Click on create
- Fill details:
 - Resource Group: gowtham-rg (select your resource group)
 - Region: West US
 - Name: any
 - Public IP address: AWS VPN public ip
- Open downloaded Note pad and search tunnel-group xx.xx.xx.xx type ipsec-l2l
- The ip address is aws vpn ip
 - Address Space: AWS VPC private address range
- From AWS select VPC -> details -> copy IPv4 ICDR range
- Review and create -> create



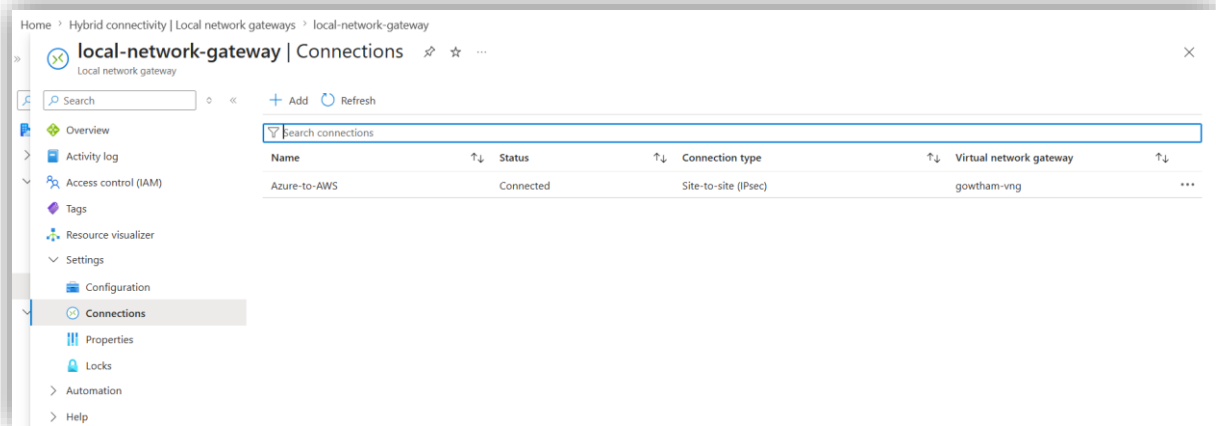
- Now open VPN gateway from Azure
- Select VPN -> settings -> Connections
- Select add
- Basic:
 - Resource Group: gowtham-rg (select yours)
 - VPN Type: Site to site
 - Name: any
 - Region: West US
- Settings:
 - Virtual Network Gateway: select existing gateway
 - Local Network Gateway: select existing gateway
 - Authentication Method: shared key

Azure–AWS Site-to-Site VPN Configuration

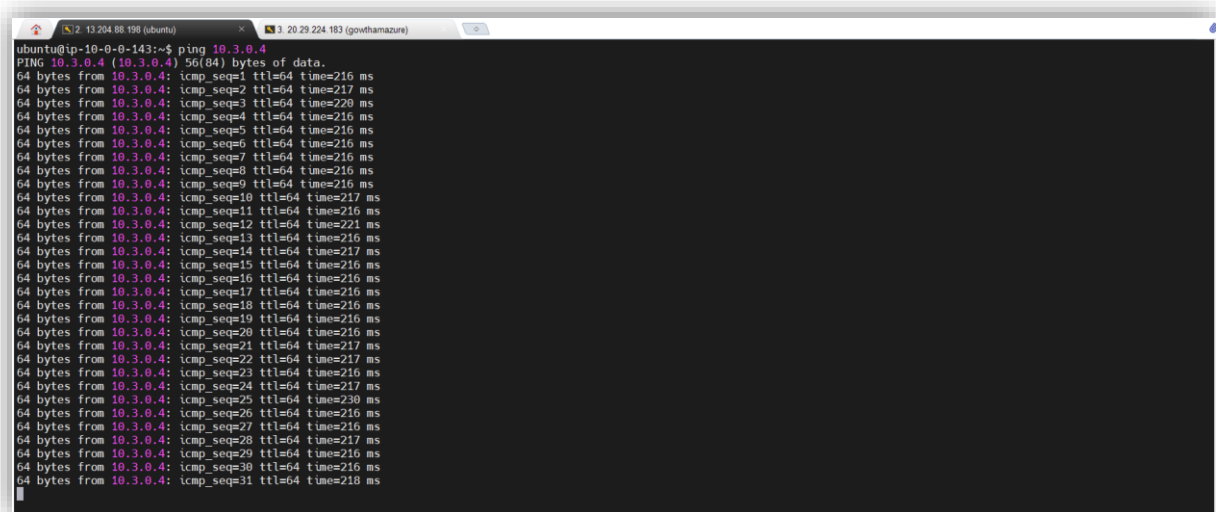
- Shared Key:

Find in notepad: search pre-shared-key that is key

- Next
- Review and Create -> Create
- Connection created but status is pending
- Wait until the status becomes connected, it will take 10 to 15 minutes to create
- Now status is connected

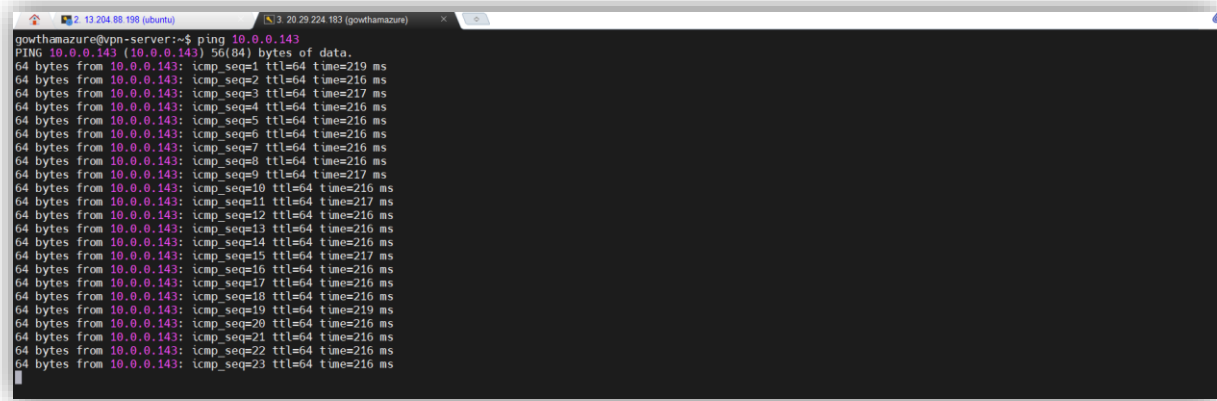


➔ After the connection Successful, now test the connection.
From AWS to Azure



From Azure to AWS

Azure–AWS Site-to-Site VPN Configuration



```
gowthamazure@vpn-server:~$ ping 10.0.0.143
PING 10.0.0.143 (10.0.0.143) 56(84) bytes of data:
64 bytes from 10.0.0.143: icmp_seq=1 ttl=64 time=219 ms
64 bytes from 10.0.0.143: icmp_seq=2 ttl=64 time=216 ms
64 bytes from 10.0.0.143: icmp_seq=3 ttl=64 time=217 ms
64 bytes from 10.0.0.143: icmp_seq=4 ttl=64 time=216 ms
64 bytes from 10.0.0.143: icmp_seq=5 ttl=64 time=216 ms
64 bytes from 10.0.0.143: icmp_seq=6 ttl=64 time=216 ms
64 bytes from 10.0.0.143: icmp_seq=7 ttl=64 time=216 ms
64 bytes from 10.0.0.143: icmp_seq=8 ttl=64 time=216 ms
64 bytes from 10.0.0.143: icmp_seq=9 ttl=64 time=217 ms
64 bytes from 10.0.0.143: icmp_seq=10 ttl=64 time=216 ms
64 bytes from 10.0.0.143: icmp_seq=11 ttl=64 time=217 ms
64 bytes from 10.0.0.143: icmp_seq=12 ttl=64 time=216 ms
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64 bytes from 10.0.0.143: icmp_seq=21 ttl=64 time=216 ms
64 bytes from 10.0.0.143: icmp_seq=22 ttl=64 time=216 ms
64 bytes from 10.0.0.143: icmp_seq=23 ttl=64 time=216 ms
```

Conclusion

This project successfully established a secure Site-to-Site VPN connection between **Microsoft Azure** and **Amazon Web Services**. The VPN tunnel enabled secure communication and data transfer between cloud resources using private networks. The project demonstrates an effective multi-cloud networking solution that enhances security, connectivity, and reliability across different cloud platforms.